

Jammula Sainath Reddy

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Product Development Intern

Analytical and results-driven Formula SAE Student Business Plan Lead with a strong background in industrial engineering, specializing in designing, fabricating, and testing formula-style racing cars.

Proficient in business plan development, encompassing technical, marketing, and financial strategies. Skilled in mechanical design and fabrication, gained during a one-month internship at Space Technology and Aeronautical Rocketry. Proficiency in CST Microwave Studio, GMAT, and PowerPoint, along with various mechanical and industrial engineering software/tools. Dedicated to applying engineering principles to tackle complex challenges in aerospace, automotive, and manufacturing. Enthusiastic about collaboration and contributing to engineering innovation.

Areas of Expertise

- Product Designing & Development
- Mechanical Design & Fabrication
- Engineering Principles & Methods
- Business Planning & Development
- Business Process Improvement
- Problem Solver & Decision Maker
- Design Concepts & Principles
- Project Management & Delivery
- Team Development & Leadership
- Effective Communication Skills
- Risk Assessment & Mitigation
- Marketing & Financial Strategies

Career Experience

Mechanical Design & Fabrication Intern | Space Technology and Aeronautical Rocketry, Surat, Gujarat, India

2022

Created detailed mechanical designs and drawings for components or systems used in rockets, spacecraft, or other aerospace technologies. Built and tested prototypes of mechanical components or systems to validate design concepts and identify areas for improvement. Supported the manufacturing process by collaborating with machinists, technicians, and engineers to ensure designs are accurately translated into physical components.

- Designed and built a 1U CubeSat at Space Technology and Aeronautical Rocketry, Surat in 2022
- Identified and troubleshooted design & manufacturing issues, proposing and implementing solutions as necessary.
- Conducted research on emerging technologies and materials that enhance systems performance and efficiency.

Formula SAE Student Business Plan Lead | Force Racing

2021 – 2022

Led and efficiently managed the business side of the Formula SAE team, including providing guidance to team members and setting goals. Conducted market research to better understand the needs and preferences of your target audience and to assess the market's potential for your team's products or services.

Vehicle Dynamics Engineer | Force Racing

2019 – 2021

Designed and analyzed the suspension system to optimize the car's handling, ride comfort, and overall performance, including selecting and tuning components such as springs, dampers, and anti-roll bars. Examined and analyzed tire performance to select the most suitable tires for the racing conditions and develop tire models for simulations. Create mathematical models and simulations of the car's dynamic behavior to understand and predict vehicle performance under various conditions. Collected and analyzed data from tests and simulations to identify areas for improvement in the car's handling and performance.

Key Projects

Academic Major project on torque vectoring.

Academic Mini Project on Adaptive headlights in automobiles which reduces the glare in oncoming vehicles and reduces the risk of accidents at night.

National Electric Kart Championship – 2023 [Captain]

SAE India Bicycle Design Competition – 2022 [Captain]

Design Of Formula Student Vehicle at Formula Bharat (Virtual) - Vehicle Dynamics Engineer & Business plan member

Design Of Formula Student Vehicle at Formula Bharat (Virtual) - Technical Manager & Business Plan Lead

Design And Manufacturing Electric Kart at NEKC-2020 [Design member]

Participated in the “National Students Space Challenge” -2019 held at IIT Kharagpur

Education

B. Tech, Mech. Engineering – Kakatiya Institute of Technology & Science: Warangal, 2023

Intermediate – Sri Chaitanya Junior College, Hyderabad, 2019

S.S.C – S. P.R High School, Hanamkonda, 2017

Licenses & Certifications

NPTEL Stars “Believer” certificate by IIT Madras

NPTEL online Elite certification on “Management Accounting” by IIT Roorkee

NPTEL online Elite certification on “Automation in Manufacturing” by IIT Guwahati

NPTEL online Elite certification on “Physics of Renewable Energy Resources” by IIT Kharagpur

NPTELonlineElite certification on “Refrigeration and Air-Conditioning” by IIT Roorkee

Honors & Awards

Rev-It! 2021- A virtual racing challenge by Formula Bharat Semifinalists

Our team was awarded 1st in the Light Weight Category at NEKC-2020

National Science Olympiad School level winner 2016

School-level Science Fair winner in 2015

Technical Proficiencies

Programming language/ software – MATLAB ONRAMP

Design Modelling software – CATIA | Fusion 360

Computational Analysis software – ANSYS | Fusion 360

Languages

English | Hindi | Telugu – Fluent